

SVM

Support vector machine.

۱. هرگاه که ما در یک دسته باشیم!

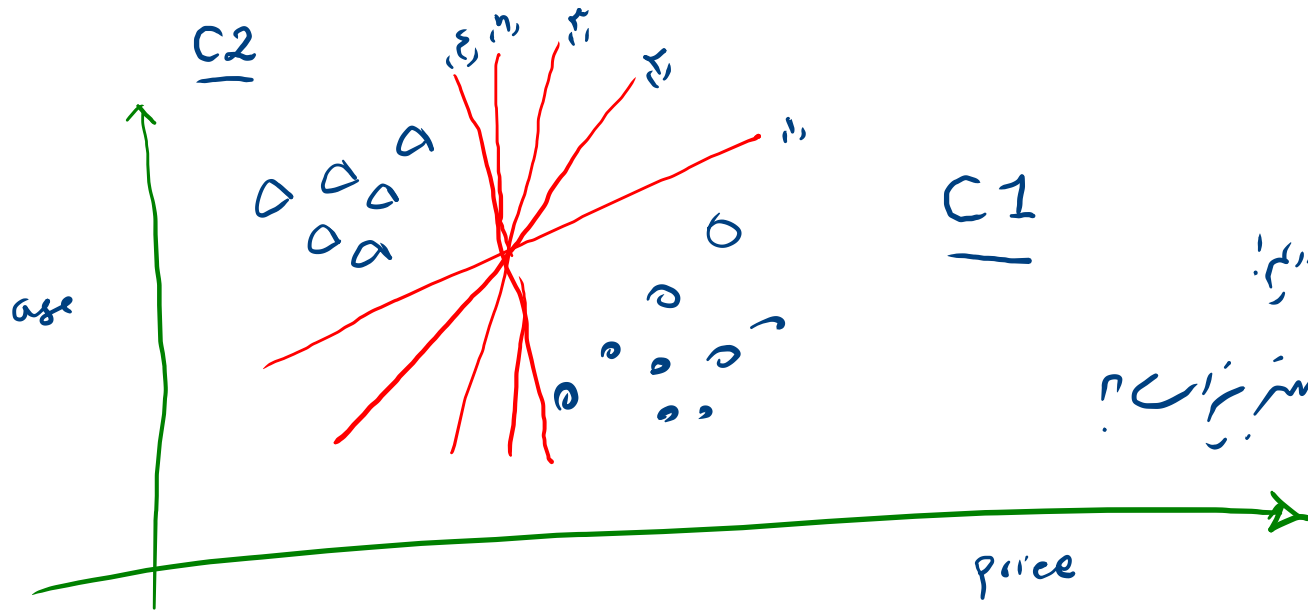
a

۲. هرگاه که، بیایم سر class، یک سر را در دستیم!

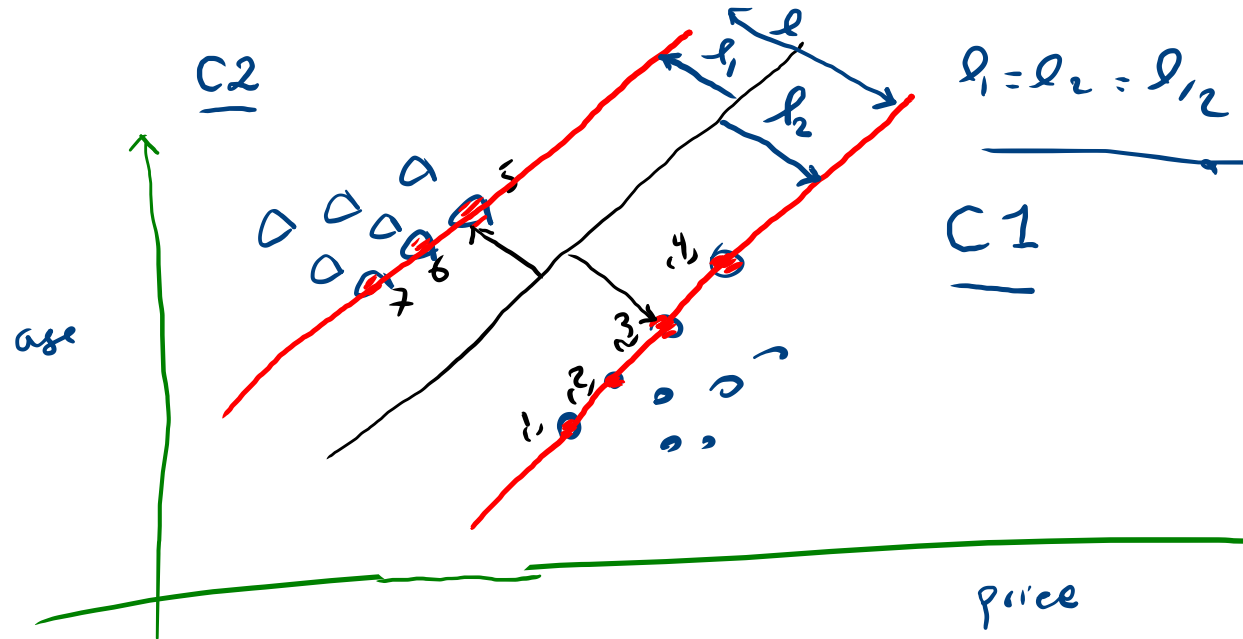
a

Sum → classification.

→ SVR: support vector Reg-



ما به نایب عدد سفید داریم!
چون که از عدد سفید ها کمتر است!



این خط‌ها خط‌های جدا کننده هستند
 هر یک از این خط‌ها خط‌های جدا کننده هستند
 خط‌های جدا کننده
 بهترین حالت را داشته باشد!

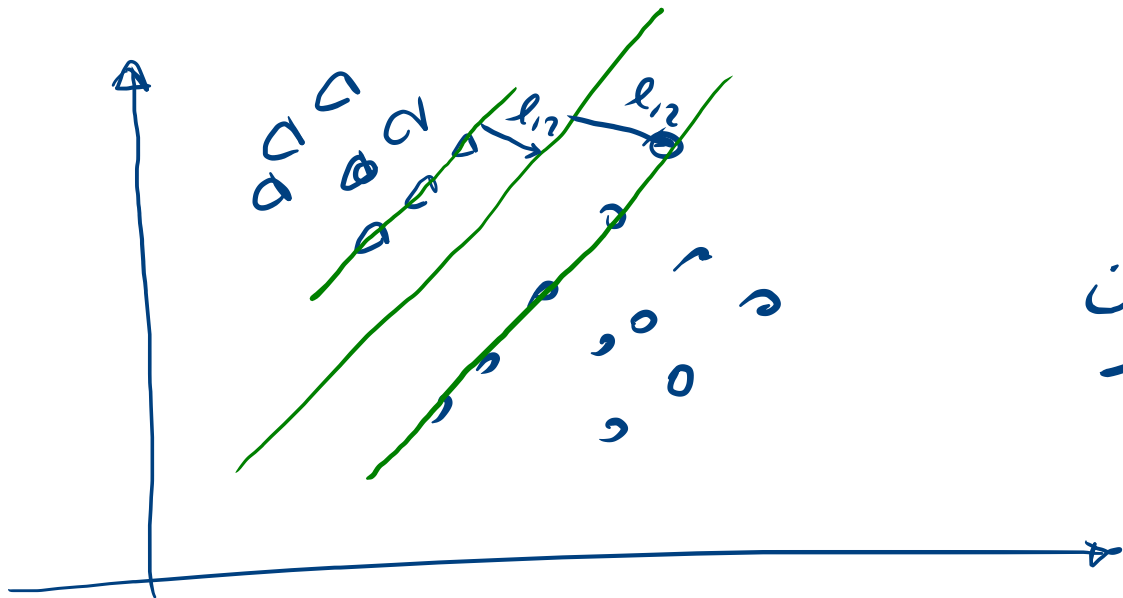
Support vectors.

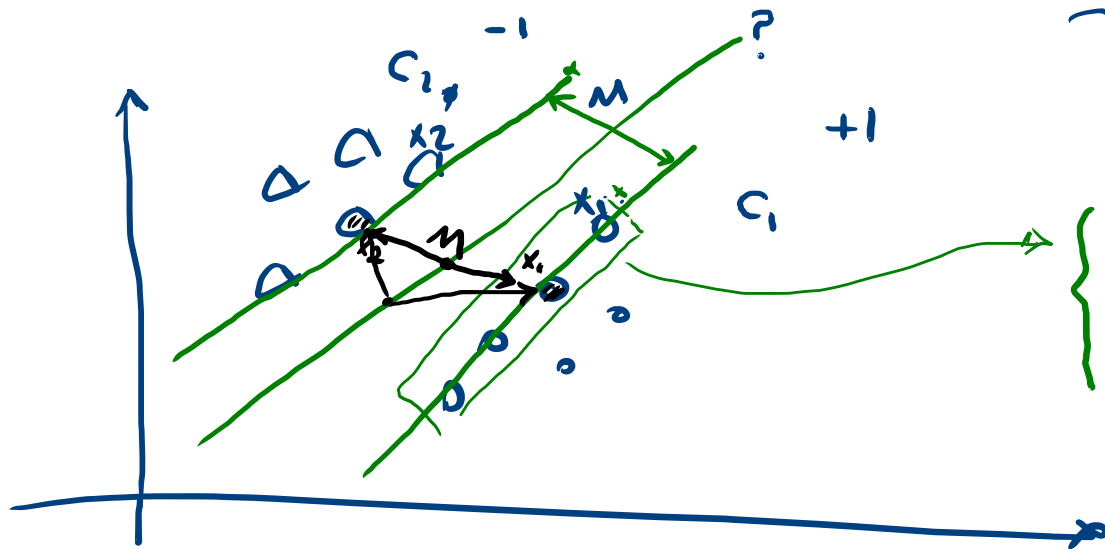
$$l_1 + l_2 \rightarrow \max$$

$$l_1 = l_2$$

08

لحسن



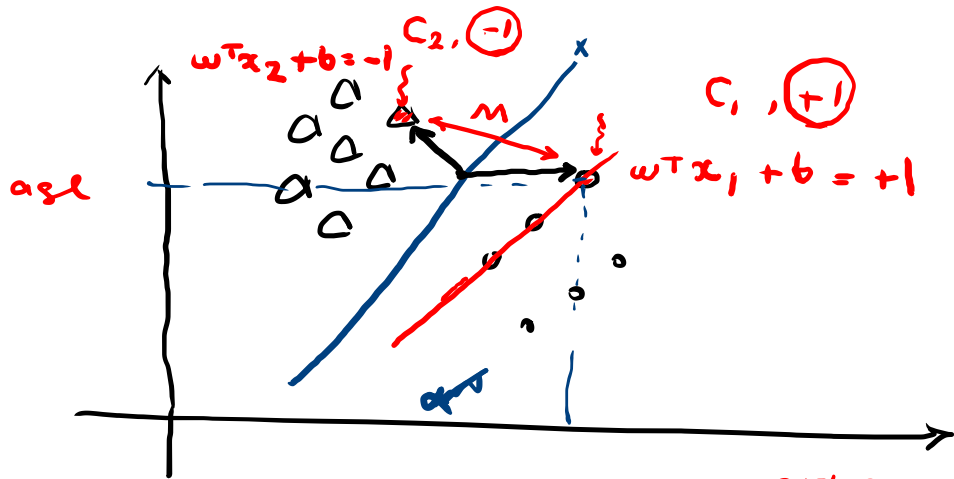


$$\begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix} \begin{bmatrix} x_1 & x_2 & \dots \\ \vdots & \vdots & \vdots \end{bmatrix} \xrightarrow{w^T x} [w_1 \ w_2 \ \dots] \begin{bmatrix} x_1 & x_2 & \dots \\ \vdots & \vdots & \vdots \end{bmatrix}$$

$$\begin{cases} +1 \rightarrow w^T x + b = +1 \\ -1 \rightarrow w^T x + b = -1 \end{cases}$$

$\swarrow$ SV_{C_1} $\nwarrow$ SV_{C_2}

C_1 $\swarrow$ SV_{C_1}
 C_2 $\swarrow$ SV_{C_2}



2D

$$w^T x_H$$

$$[w_1 \ w_2] \begin{bmatrix} price \\ ask \end{bmatrix}$$

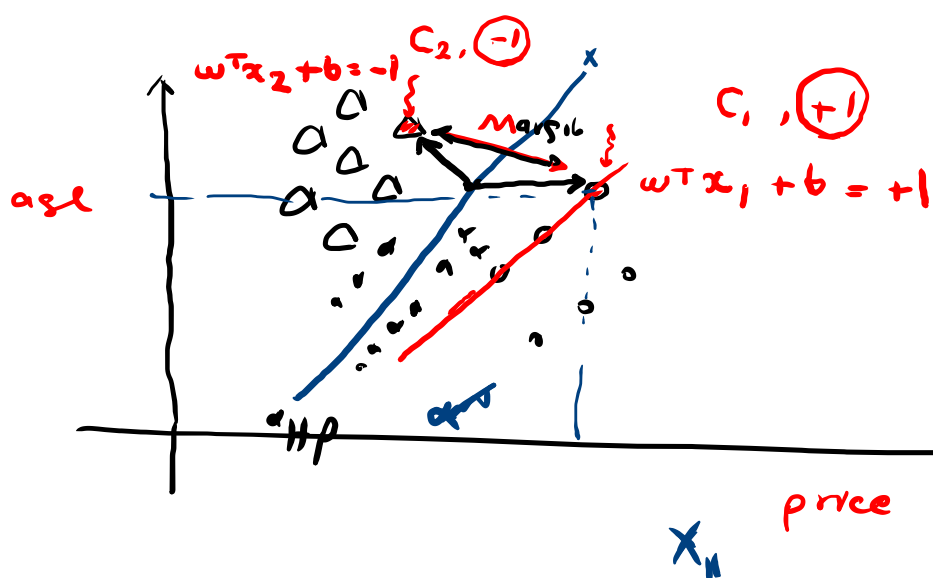
$$C_1: \quad w^T x_1 + b = +1$$

$$C_2: \quad w^T x_2 + b = -1$$



$$a \quad \underline{w^T x + b = 0} \quad \text{Hyperplane.}$$

$$\left[\begin{array}{l} \text{Feature space} \rightarrow 3d \rightarrow \text{HP: } 2d \\ \text{"} \rightarrow 4d \rightarrow \text{HP: } 3d \\ \text{"} \rightarrow 100d \rightarrow \text{HP: } 99d \end{array} \right]$$



$$C_1: \omega^T x_1 + b = +1 \rightarrow \text{SVC}_1$$

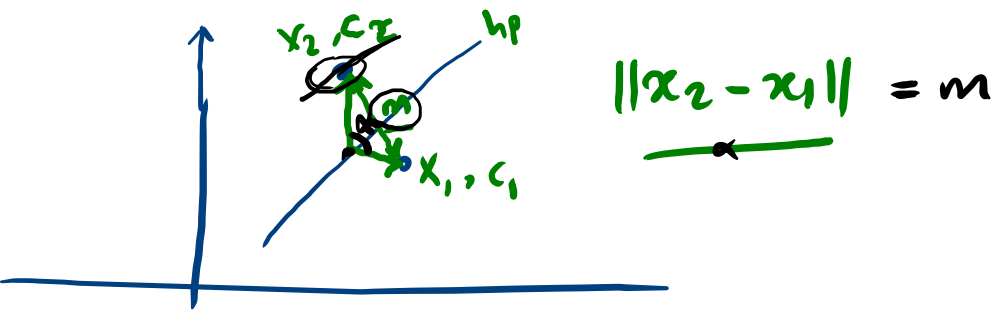
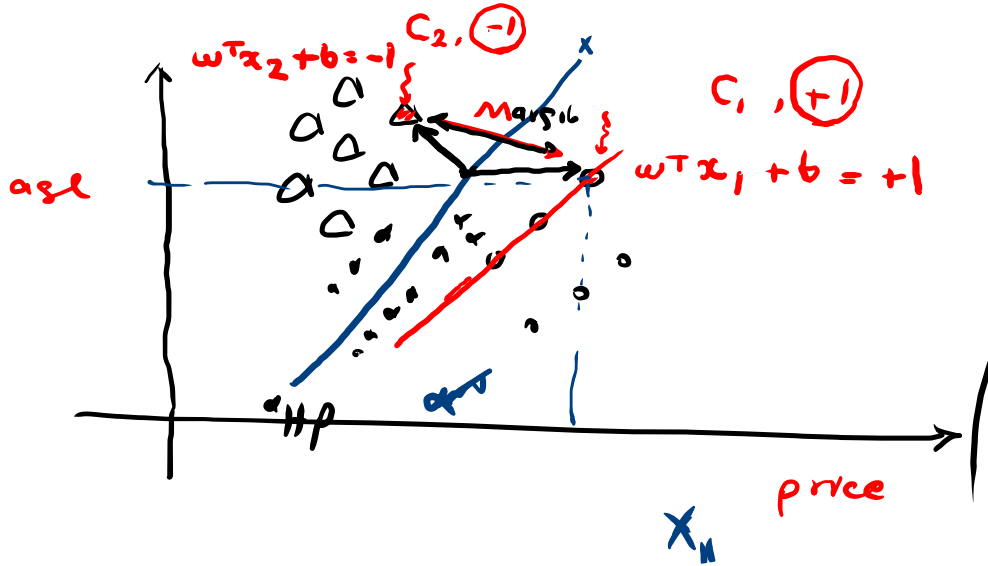
$$C_2: \omega^T x_2 + b = -1 \rightarrow \text{SVC}_2$$

$$\omega^T x + b = 0 \quad \text{Hyperplane.}$$

other points

$$\begin{cases} y^+ (\omega^T x_1 + b) \geq 1 \\ y^- (\omega^T x_2 + b) \geq 1 \end{cases} \quad \leftarrow \quad \begin{cases} C_1 \\ C_2 \end{cases} \quad \begin{cases} \omega^T x_1 + b \geq 1 \rightarrow y^+ = \textcircled{1} \\ \omega^T x_2 + b \leq -1 \rightarrow y^- = -1 \end{cases}$$

$$\boxed{y (\omega^T x + b) \geq 1}$$



$$\underbrace{\|w\| \|x_2 - x_1\| \cos \alpha}_m = 2$$

$$c_1 \rightarrow w^T x_1 + b \geq 1, y^+ = +1$$

$$c_2 \rightarrow w^T x_2 + b \leq -1, y^- = -1$$

$$\boxed{y(w^T x + b) \geq 1} \quad (1)$$

$$\begin{cases} w^T x_2 + b = -1 \\ w^T x_1 + b = +1 \end{cases}$$

$$w^T \cdot (x_2 - x_1) = 2$$

$$\boxed{w^T \cdot (x_2 - x_1) = 2} \quad (2)$$

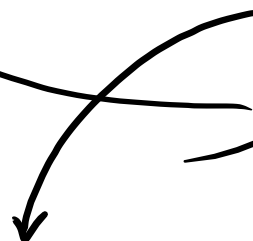
$$m = 2 \text{ آف } 11$$



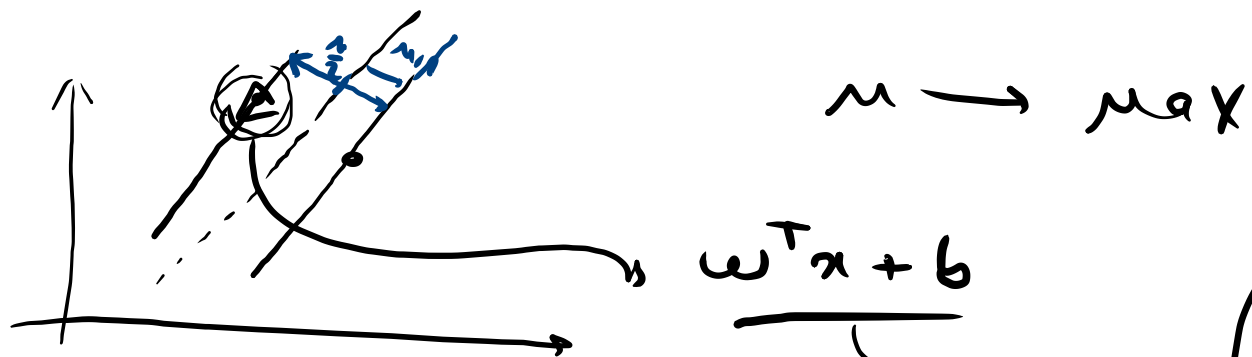
$$m = \frac{2}{11311}$$

کار

Sum ایش m کار



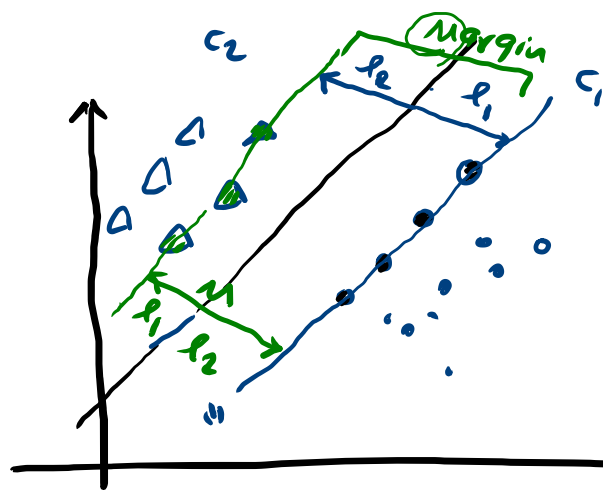
بجای ۲ کمتر هست!



$$\omega^T x + b$$



دشمن به جمع خود را جمع کند
و ما را منفرد کند!



$$\underline{l_1 + l_2 = \underline{\underline{max}}}$$

هدف از سبب زماص از حد است!، ارادته است. (l_1)
 و همین سبب زماص از حد است!، ارادته است. (l_2)

$$\underline{M = l_1 + l_2}$$

$$l_1 = l_2 = M/2$$

$$l_1 = l_2 = l_{1,2}$$

$$c_1: +1$$

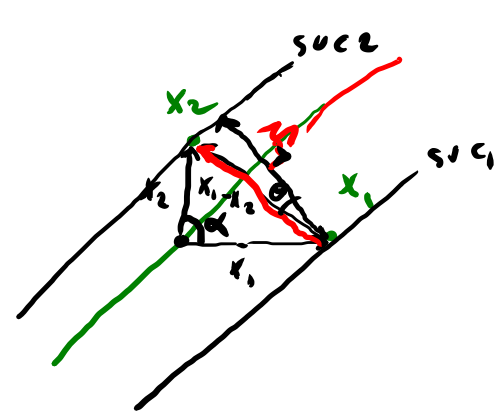
$$c_2: -1$$

$$y(x_1) = +1$$

$$y(x_2) = -1$$

$$\left. \begin{array}{l} \text{su} \left\{ \begin{array}{l} \omega^T x_1 + b = +1 \\ \ominus \\ \omega^T x_2 + b = -1 \end{array} \right. \begin{array}{l} c_1 \\ c_2 \end{array} \end{array} \right\}$$

$$\omega^T \cdot (x_1 - x_2) = 2$$



$$\left[m = \|x_1 - x_2\| \cos \theta \right]$$

$$\left[\omega^T (x_1 - x_2) = 2 \rightarrow \underbrace{\|\omega^T\| \|x_1 - x_2\| \cos \alpha}_{m} = 2 \right]$$

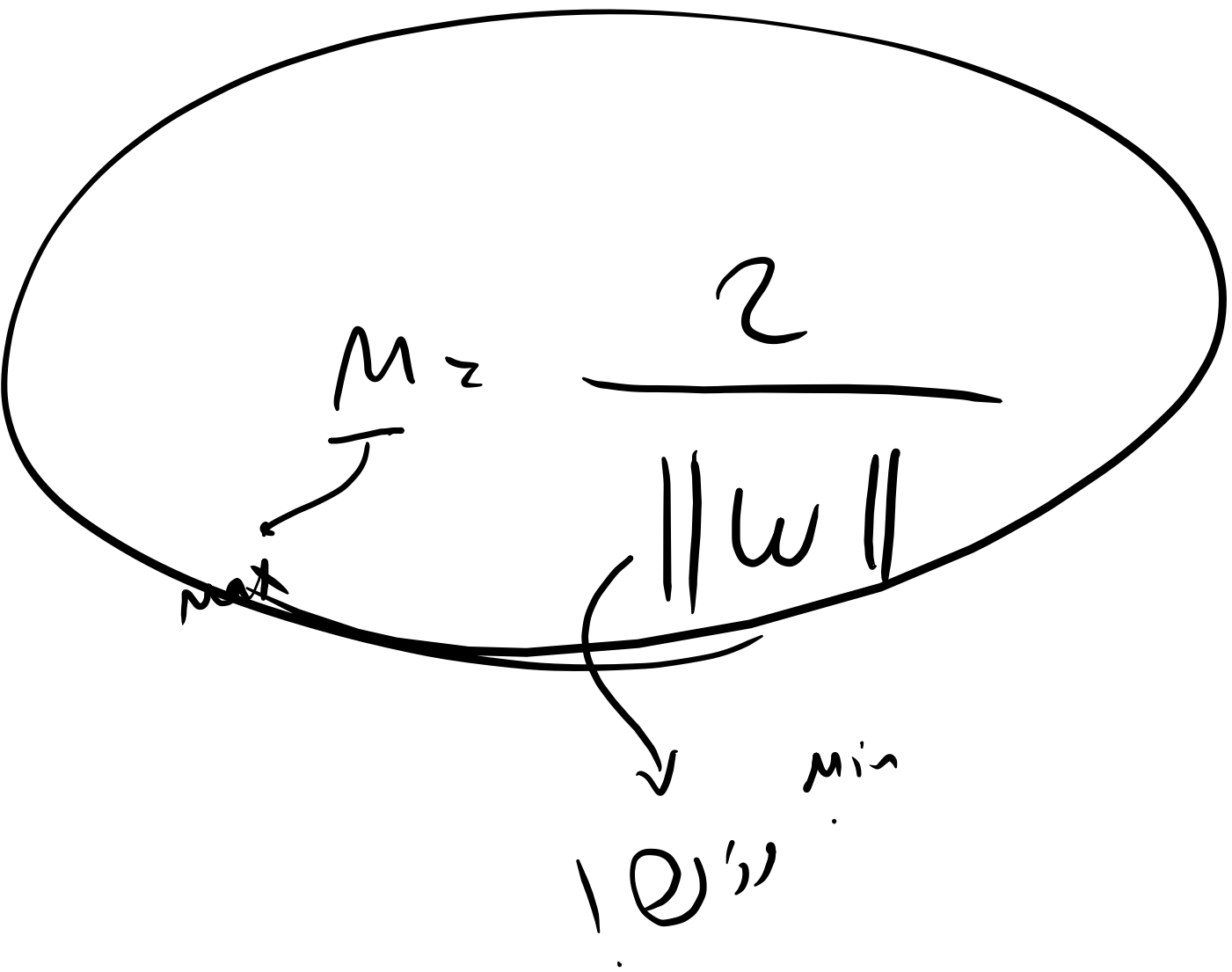
$$\|x_1 - x_2\| \cos \theta = \|x_1 - x_2\| \cos \alpha$$

? $\cos \theta = \cos \alpha$

$(x_1 - x_2)^T = \overline{m}$

$\|\omega^T\| = \|\omega\|$

$m = \frac{2}{\|\omega\|}$



The End

